



Many equations, one solution

- 1 This bar model represents the initial equation $2x + 3 = 9$. What needs to be done to the top bar to maintain equality? Tick **3** correct answers



- ☒ + 9
☐ + $2x$
☒ + $(2x + 3)$
☐ $\times 1$
☒ $\times 2$

- 2 Which of these calculations show '6 plus 3 all multiplied by 2'? Tick **2** correct answers

- ☐ $6 + 3 \times 2$
☒ $2(6 + 3)$
☐ $2 \times 6 + 3$
☒ $2 \times 6 + 2 \times 3$
☐ $6 + 3 + 6$

- 3 Lucas manipulates the equation $2x + 8 = 6$. Which of these equations could he obtain and maintain equality? Tick **3** correct answers

- ☒ $x + 4 = 3$
☒ $2x + 10 = 8$
☐ $10x + 8 = 30$
☐ $2x = 6$
☒ $4x + 8 = 2x + 6$



- 4 Which expression could fill the blank on the right hand side of this equation to maintain equality? Tick **3** correct answers

$$\begin{array}{rcl}
 2x + 4 & = & 6 \\
 \underline{\hspace{2cm}} & = & 3
 \end{array}$$

☒ $2x + 1$

☐ $2x + 2$

☐ $x + 4$

☒ $\frac{2x + 4}{2}$

☒ $x + 2$

- 5 Which of these equations has a solution of $x = 6$? Tick **2** correct answers

- ☐ $x + 5 = -1$
- ☒ $2x - 3 = 9$
- ☒ $5 + 3x = 2x + 11$
- ☐ $10 - x = 16$
- ☐ $15 - 2x = 2x$

- 6 Which of these is a solution to the equation $18 - 6x = 12 - 4x$? Tick **1** correct answer

- ☐ $x = -4$
- ☐ $x = -1$
- ☐ $x = 0$
- ☐ $x = 2$
- ☒ $x = 3$

